

Groundwater

Issue Paper/

Managing Groundwater to Ensure Ecosystem Function

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Abstract

Groundwater is a critical resource not only for human communities but also for many terrestrial, riparian, and aquatic ecosystems and species. Yet groundwater planning and management decisions frequently ignore or inadequately address the needs of these natural systems. As a consequence, ecosystems dependent on groundwater have been threatened, degraded, or eliminated, especially in arid regions. There is growing acknowledgment that governmental protections for these ecological resources are necessary, but current legal, regulatory and voluntary provisions are often inadequate. Groundwater management premised on “safe yield,” which aims to balance human withdrawals with natural recharge rates, typically provides little to no consideration for water needed by ecosystems. Alternatively, the “sustainable yield” concept aims to integrate social, economic and environmental needs for groundwater, but the complexity of groundwater systems creates substantial uncertainty about the impact that current or future groundwater withdrawals will have on ecosystems. Regardless of the legal or regulatory framework, guidance is needed to help ensure environmental water needs will be met, especially in the face of pressure to increase human uses of groundwater resources. In this paper, we describe minimum provisions for planning, managing, and monitoring groundwater that collectively can lower the risk of harm to groundwater-dependent ecosystems and species, with a special emphasis on arid systems, where ecosystems and species may be especially reliant upon and sensitive to groundwater dynamics.

Introduction

Human reliance on groundwater is increasing worldwide, driven in part by fully subscribed or overallocated surface water systems (Grantham and Viers 2014)

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and the direct and indirect effects of climate change (de Graaf et al. 2019). Groundwater is particularly important in arid regions with limited surface water supplies. It is present almost everywhere under the land surface (Alley et al. 1999), but unlike surface water, groundwater flows and connectivity are often not easily measured or known. Groundwater laws, regulations and policies exist in many jurisdictions across the United States and globally (Groundwater Governance Project 2016; Nelson and Ouevaullier 2016), but even where they exist, only rarely do they consider—and require current and future users to avoid—impacts to natural systems (Megdal et al. 2015; Rohde et al. 2017).

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Access to groundwater is essential for the viability of many terrestrial, riparian, and aquatic ecosystems and associated plant and wildlife species (Naumburg et al. 2005; Kath et al. 2018). Groundwater-dependent streams, wetlands, springs, seeps, shrublands and forests are increasingly threatened by the cumulative impacts of past and current groundwater withdrawals (Rohde et al. 2019). Lagged aquifer responses to historic groundwater withdrawals, ongoing uses and new water users, and variability in recharge rates, including due to a changing climate, can all lead to water shortages for ecosystems (Alley et al. 1999; Reilly et al. 2008; Gleeson et al. 2012). Degraded or destroyed natural systems can adversely affect human communities in many ways, including the loss of important ecosystem services (McCarl et al. 1999; Gungle et al. 2016). Environmental water needs have had a history of neglect in the planning and management of groundwater resources and withdrawal projects, partly due to scientific knowledge gaps on where groundwater-dependent ecosystems exist and how groundwater dynamics affect their viability.

Historically, “safe yield” has been the paradigm for managing groundwater resources, which is premised on balancing human withdrawals with natural recharge rates (Bredehoeft et al. 1982; Bredehoeft 2002; Alley and Leake 2004; Gungle et al. 2016). For a variety of reasons, this approach essentially ensures that natural systems will receive little to no groundwater over the long term (Alley and Leake 2004; Kalf and Wooley 2005; Zhou 2009; Richter et al. 2014; Gungle et al. 2016). In contrast, the “sustainable yield” concept aims to balance the water needs of both natural and human communities. Alley et al. (1999) define groundwater sustainability as the “development and use of groundwater in a manner that can be maintained for an indefinite time without causing unacceptable environmental, economic, or social consequences.” Achieving groundwater sustainability requires addressing the temporal and spatial dynamics of groundwater systems, groundwater withdrawals, and associated consequences, including impacts to natural systems (Alley et al. 1999; Rudestam and Langridge 2014).

Challenges of Managing Groundwater to Ensure Ecosystem Function

Applying sustainable yield principles to protect ecosystems is challenging for several reasons. First, groundwater systems are spatially and temporally complex, leading to considerable uncertainty of groundwater flow dynamics (Reilly et al. 2008). The nature of groundwater movement, dynamics, and interactions with natural systems can cause effects to take years, decades or centuries to be seen, and even longer to reverse (Eamus et al. 2006; Barlow and Leake 2012). Some effects, such as land subsidence or species extinction may not be reversible (Konikow 2015; Wolaver et al. 2020). Managing for sustainability becomes even more challenging when the distance between groundwater withdrawals and impacted systems is expansive and crosses jurisdictions. Such interconnections and lags can complicate

disentangling impacts of a given project from external effects like climate change (Gregory et al. 2006).

Second, groundwater-dependent ecosystems are often associated with endemic and sensitive species with low resilience to impacts (Sada et al. 2005; Kodric-Brown and Brown 2007; Freed et al. 2019; Wolaver et al. 2020). In arid regions in particular, such species may be found in only a few locations (Wolaver et al. 2020). Pumping even small quantities may cause springs to dry, which can eliminate species with perennial water needs (Sada et al. 2005; Museum of Northern Arizona Springs Stewardship Institute 2020; Wolaver et al. 2020).

Third, few locations have sufficient available empirical datasets with which to adequately characterize the groundwater system and associated natural systems. Refinement and calibration of numerical groundwater models require data about spatial and temporal distribution of recharge, vertical and hydraulic head gradients, interaction between the aquifer and surface waters, hydrogeology, and evapotranspiration rates (Pool and Dickinson 2007). Lack of data contributes to model parameter uncertainty associated with groundwater flow dynamics (Reilly et al. 2008). Ecological dynamics can have even greater parameter uncertainty because of limited knowledge of ecological processes (Kath et al. 2018), which further challenges the development of accurate models.

Finally, stakeholder and institutional support for successful implementation of groundwater sustainability principles requires considerable time, funding and effort. The long duration and high uncertainty associated with sustainable groundwater management means that institutional capacity and commitment to ensure successful protection of natural systems must be sustained (Richter et al. 2014; Thomann et al. 2020). The costs of installing and maintaining the infrastructure and technical capacity required to collect and analyze extensive empirical data may be seen as cost-prohibitive and will likely be difficult to guarantee over the long term, especially if many stakeholders with different perspectives are involved (Gregory et al. 2006).

A Need for Guidance That Protects Ecosystems

In the arid western United States, groundwater use is widespread and increasing (Perrone and Jasechko 2017), with added pressure due to drought and projected climate change impacts (Babbitt et al. 2018; Dieter et al. 2018). For example, groundwater withdrawals for irrigation in California increased 60% from 2010 to 2015, a period of protracted drought; surface water withdrawals for irrigation decreased 64% during the same period (Dieter et al. 2018). Partly in response to this drought, the State of California enacted the Sustainable Groundwater Management Act of 2014, which requires that impacts to groundwater-dependent ecosystems be considered and reported in groundwater sustainability plans (Rohde et al. 2017). However, legal frameworks for planning and managing groundwater use vary greatly between U.S. states and even between areas within a single state (Nelson and Ouevaullier 2016). California’s Sustainable

Groundwater Management Act only applies in some of the state's groundwater basins (Parker et al. 2021). Similarly in Arizona, some areas within the state are highly managed while other areas remain largely unregulated (Arizona Department of Water Resources 2021).

Major groundwater development projects have also been proposed (e.g., Southern Nevada Water Authority Groundwater Development Project [Deacon et al. 2007]; Cadiz Conservation, Recovery and Storage Project [Sizek 2018; Zdon et al. 2018]). In such large volume projects, even small inaccuracies in estimated groundwater dynamics can lead to errors in groundwater model predictions that can result in major, even catastrophic, impacts on natural systems. Moreover, once the uses of the groundwater are established for such large volume projects, societal forces would likely resist curtailment of withdrawals, even if adverse ecological impacts are being observed, especially if project financing is dependent on continued extraction. For these reasons, the opportunity for precaution is prior to a use being locked in via project approval (Cantarelli et al. 2010; Flyvbjerg 2014).

Given the sensitivity of natural systems to reduced groundwater availability, particularly in arid areas (Patten et al. 2008; Davis et al. 2017), and the variability among regulatory frameworks (Nelson and Ouevauller 2016), guidance is needed for planning, managing and monitoring groundwater systems and projects to ensure groundwater sustainability, inclusive of environmental water needs. Here, we provide that guidance. Our framework is based on sustainable yield principles, and takes into account the hydrogeology of the system, the connection of the ecological system to groundwater dynamics, and the many uncertainties that are present due to historic, current and future conditions, with special attention to arid regions because those ecosystems and species may be especially reliant upon and sensitive to groundwater dynamics. We discuss why these provisions collectively are necessary in order to plan, monitor and adaptively manage groundwater use sustainably. We also caution that further study will be required to determine the sufficiency of these provisions for protecting groundwater-dependent ecosystems and species.

Minimum Provisions for Protecting Natural Systems and Maintaining Groundwater Sustainability

Managing groundwater sustainably requires that the combined effects of historic, current and proposed water use not exceed hydrologic thresholds needed to ensure the continued function of groundwater-dependent ecosystems. We describe seven provisions necessary to incorporate into basin management or project planning to reduce the risk of adverse impacts: (1) *define specific groundwater management goals, objectives, and thresholds* for existing human and natural communities within a given basin or aquifer; (2) *define clear triggers for management actions* that are science-based and protective of natural systems; (3) *use predictive models* to assess the

relative performance of various management options and mitigation alternatives, identify monitoring networks and locations, and set thresholds for adverse impacts and triggers to avoid impacts before they occur; (4) *implement a robust monitoring plan* at appropriate temporal and spatial scales; (5) *provide accessible and timely reporting* of data and model results; (6) *implement management actions* to avoid, minimize, and mitigate impacts in a timely manner; and (7) *secure adequate funding* and capacity for planning, modeling, management, and monitoring to accomplish sustainable groundwater use in terms of the environmental, social and economic aspects. These provisions largely tailor principles of the mitigation hierarchy to this application, that is, to first avoid, then minimize, and lastly mitigate any remaining impacts of groundwater use on natural systems (Kiesecker et al. 2010; McKenney and Wilkinson 2015). They are also premised on the precautionary principle, in which a decision-maker must consider the threat of irreversible damage as a reality (Slattery 2016) and provide for protection against this harm even if existing science is not certain (Crawford-Brown and Crawford-Brown 2011).

Minimum Provision 1: Clearly Define Management Goals and Objectives with Thresholds to Avoid Adverse Impacts to Natural Systems

Groundwater management goals, objectives, and thresholds that reflect the unique combination of water needs for individual species, natural communities, and existing and future human communities of a given location are critical to ensure prediction and prevention of adverse conditions (Christian-Smith and Abhold 2015; Fischman and Ruhl 2016). As used here, *goals* set out the long-term desirable future conditions, *objectives* refer to measurable targets to achieve the goal(s), and *thresholds* define unacceptable or undesirable states related to those measurable targets (Meadows 1998; Gleeson et al. 2012; Christian-Smith and Abhold 2015). Management goals and thresholds are customarily set by regulatory or permitting agencies in accordance with prevailing legal requirements (Groundwater Governance Project 2016), but sometimes arise in local collaborative planning or other contexts. For example, California's Sustainable Groundwater Management Act requires certain groundwater basins to establish a sustainable goal, sustainable yield, and measurable objectives and thresholds during groundwater management planning (Christian-Smith and Abhold 2015; Newman et al. 2018), whereas a sustainable yield goal was adopted locally for the Sierra Vista Subwatershed in Arizona through a collaborative response of 21 stakeholders to Section 321 of the National Defense Authorization Act (Upper San Pedro Partnership 2013).

Sustainable groundwater planning and management should begin by setting a holistic basin management goal using a multi-generational time horizon (50 to 100 years or longer) that defines the desired future condition (Gleeson et al. 2012). Management objectives, policies

and programs will be developed based on achieving the management goal. Consequently, goals that are not defined at the appropriate scope (e.g., encompassing the relevant geographic area, planning horizon, and community and ecological values) can lead to inadequate or misleading indicators, and management programs that are ill-equipped to handle unintended or undesired results (Meadows 1998; Groves et al. 2019). For example, the Phoenix Active Management Area in Arizona encompasses several sub-basins; although it has a safe yield goal that is being met, some sub-basins nevertheless experience localized declines in groundwater levels (Babbitt et al. 2018). Being explicit about goals and the values and assumptions underlying them is critical for defining and measuring success, identifying the best strategies to achieve them, and dealing with uncertainties and changing conditions in complex systems (Groves et al. 2019).

Objectives to achieve the goal should be established using appropriate spatial and temporal scales and incorporate hydrologic, economic, and environmental considerations. Gleeson et al. (2012) recommend “backcasting” be used to develop specific policies and programs needed to achieve the goal by working backwards from the desired future to the present. Christian-Smith and Abhold (2015) and Slattery (2016) further point out that objectives should be measurable so that it is possible to track progress toward management goals. Under California’s Sustainable Groundwater Management Act, spatial and temporal objectives and thresholds are required for six management criteria (i.e., groundwater levels, groundwater storage, groundwater quality, seawater intrusion, land subsidence, and surface water depletions) to ensure that adverse impacts do not occur to groundwater users, including the environment (Christian-Smith and Abhold 2015).

Identifying science-based thresholds and objectives for natural systems is necessary to quantify groundwater management objectives that are protective of ecosystems. In this context, a threshold is defined as the groundwater conditions that transition an ecosystem from a stable state of structure and function to an unacceptable or undesirable state (Groffman et al. 2006; Moritz et al. 2013). Objectives for natural systems are the groundwater conditions that sustain the ecosystem in such a way that it has the physiological capacity to deal with natural variations in groundwater levels. For practical purposes, thresholds and objectives are best established for an ecological target that can indicate whether changes are occurring in the ecosystem (Rohde et al. 2020). For example, springsnails may be an appropriate ecological target for defining objectives and thresholds within a spring ecosystem since springsnails rely entirely on groundwater and would cease to exist if spring flow declined beyond a certain amount (Museum of Northern Arizona Springs Stewardship Institute 2020). For rivers, a suitable ecological target might be a riparian bird species because it is reliant on riparian forest habitat (Merritt and Bateman 2012). Since groundwater sustainability can be highly uncertain with large impacts and time lags, thresholds and objectives should be conservative and precautionary (Zipper et al. 2020).

For example, groundwater withdrawals throughout the year in Michigan are limited to volumes associated with fish community needs during the most sensitive time of year to avoid low-flow thresholds (Kendy et al. 2012). In many cases, multiple temporal and spatial ecosystem thresholds and objectives may be necessary to quantify management objectives for a given groundwater system (Slattery 2016; Gillson et al. 2019).

Unfortunately, science-based thresholds and acceptable ranges of variation for target species are not always well-defined due to lack of data or ecohydrological understanding (Eamus et al. 2006; Orellana et al. 2012). Thus, groundwater planners and project proponents may need to expend more time (e.g., years, decades) and resources (e.g., monitoring, modeling, analysis) filling this gap so that ecosystem impacts can be adequately considered. In the absence of adequate information about thresholds for these resources, assumptions about how much water can sustainably be withdrawn should be conservative to increase the margin of safety for uncertain system responses (Christian-Smith and Abhold 2015; Zipper et al. 2020).

Minimum Provision 2: Define Science-Based Protective Triggers for Action

In addition to thresholds, an important component of effective management is the incorporation of clearly defined triggers (i.e., commitments that specify when actions are needed if monitoring indicates a trend toward a threshold, even over a very long time interval; Nie and Schultz 2012) to ensure that the established threshold is neither reached nor exceeded (Christian-Smith and Abhold 2015). In the aforementioned springsnail example, a trigger to avoid reaching the flow threshold might be based on water levels monitored in a well situated between a groundwater withdrawal point and the focal spring; a corrective action (e.g., reduction in pumping) would be prescribed when levels are observed that would indicate an approach to that threshold—importantly, before the anticipated adverse impact is manifest at the focal spring. Effective triggers must be associated with appropriate monitoring in space and frequency (Noorduijn et al. 2019; see Minimum Provision 4) and require objective analyses of those data, institutional capacity empowered to take action, and an enforceable mandate to take action if triggers are reached, up to and including ceasing withdrawals (Craig and Ruhl 2014).

In arid environments, the loss of any surface or near surface water flow has the potential to irreversibly impact groundwater-dependent ecosystems (Stromberg et al. 1992; Glazer and Likens 2012). Springs and seeps are critical water sources that support a range of species and riparian, upland, and aquatic plant communities (Patten et al. 2008; Hjort et al. 2015). Thus, the design of management actions should aim to protect these vulnerable resources proactively and predictively, because even minimal drops in water levels, including those that occur incrementally over long time periods or those that are

due to pumping wells at some distance from the target resources, can extirpate species and natural communities (Deacon et al. 2007; Glazer and Likens 2012). Groundwater models can inform the design of trigger locations between withdrawal points and natural communities that can detect trends in monitoring wells or surface flows long before the lagged responses to groundwater withdrawals have spatially migrated and caused undesired ecological impacts (Eamus et al. 2006; Wolaver et al. 2020). Noorduijn et al. (2019) used a generic groundwater model and standard groundwater flow equations to demonstrate how water table declines can persist for years even after activating a trigger to cease pumping.

We underscore that triggers based solely on direct ecological monitoring of the condition of communities or species or groundwater levels at the target ecosystem are unlikely to be effective. Monitoring water levels and flows close to withdrawal points is the preferred approach because it is more likely to detect effects heading toward groundwater-dependent ecosystems well in advance of an adverse impact on those target resources (Harrington et al. 2017; Noorduijn et al. 2019).

Minimum Provision 3: Use Predictive Groundwater and Ecological Models to Prioritize the Most Effective Management Strategies

Groundwater development and associated impacts to ecosystems involve uncertainty because groundwater is underground and mostly out of sight (Reilly et al. 2008; Christian-Smith and Abhold 2015) and because of environmental and climatic variability (Alley et al. 1999). Predictive models can play a key role in the evaluation of sustainability in terms of developing strategies to have enough water at the right times and places to meet various needs, and determining how much water can be withdrawn before reaching triggers and thresholds (Alley et al. 1999; Barlow and Leake 2012; Wolaver et al. 2020). Well-designed predictive models populated with adequate data are most effective at developing mitigation actions and associated triggers (Slattery 2016).

Ecological models can be informative complements to groundwater models by linking how hydrologic change leads to ecological responses, allowing for the forecasting of adverse impacts long before they might occur, and providing predictions of water use that may be possible without surpassing triggers and thresholds. Orellana et al. (2012) provide a review of approaches to quantify groundwater use by vegetation, including modeling techniques with variably saturated and saturated groundwater models that include vegetation water use, coupled subsurface models with surface water or land surface models, and lumped parameter models. Linked ecological and hydrologic models can be informative for examining potential effects of alternative management strategies in advance of any actual impacts (Rains et al. 2004; Eamus et al. 2006).

Participatory modeling in which stakeholders are involved in conceptualization, development, validation, and improvement of models can be a means of not only generating predictions about the system of interest, but also fostering communication between participants and increased stakeholder confidence and understanding in the constructed model and outcomes (Crevier and Parrott 2019). For example, in the Upper San Pedro Partnership (USPP) in Arizona, a collaboratively-developed regional groundwater model established a common understanding of the surface and groundwater systems as well as human and natural system water needs that has enabled the USPP to begin implementing actions to sustain baseflows in the San Pedro River (Richter et al. 2014). This collaboration developed after many years of conflict stemming from water rights adjudication struggles and provisions of Section 7 of the Endangered Species Act. A collaborative approach to identify an agreed upon set of proactive management measures proved more productive than the development of multiple conflicting groundwater models. In some cases, however, such as large new water development projects where there may be a pronounced imbalance in the capacity, knowledge, and financial interests of stakeholders, the effectiveness of participatory approaches may be limited (Ostrom et al. 2007).

The accuracy of any model is only as good as the empirical data used to develop and test it (Kirchner 2006), and adequate hydrogeologic and ecological data are commonly lacking for many areas. Ecological data in particular can be limited, so connecting groundwater changes with groundwater-dependent ecosystem responses can be challenging (Kath et al. 2018). In many places, there is also limited understanding of the relationships between groundwater dynamics and natural systems (Rudestam and Langridge 2014; Kath et al. 2018). In addition to more studies to discern flow-ecology relationships, statistical modeling approaches may be useful at landscape scales for estimating these relationships (e.g., combining remotely sensed data with groundwater models or indicators; Kath et al. 2018). Sensitivity modeling and exploration of assumptions of system conceptualizations (e.g., the presence of faults or zones of high hydraulic conductivity) are recommended for siting monitoring and incorporating quantified uncertainty into protective management triggers (Kriebel et al. 2001; Slattery 2016; Thomann et al. 2020).

Although good practice for modeling involves continually updating and improving models as new data are gathered (Alley et al. 1999), groundwater or ecological models that are poorly calibrated or inaccurate should not be used to make predictions under the assumption that they will be improved over time, or accepted as a basis for proceeding to implement a project. Collecting the data required for adequate model calibration and understanding of baseline conditions is needed before any actions are implemented (Slattery 2016). We recognize that may require more time and expense than decision makers, project developers and investors may prefer. However, high volume groundwater withdrawals put

the future water security of existing human and natural communities at most risk. Thus, we recommend that proposals for large withdrawals not move forward unless the project proponents can clearly establish that management objectives and sustainability goals will be met. Even where harm appears unlikely, the development and use of a well-designed, adequately documented and calibrated groundwater model before project approval will help investigate whether additional water uses are consistent with achieving management goals and avoiding the risk of adverse social, economic, and environmental consequences over the long term (Slattery 2016; Thomann et al. 2020). This is especially important when financing for a large-scale project is dependent on water delivery, as it is often politically and economically difficult to adaptively curtail withdrawals after high-capacity facilities have been built and that water has been put to use.

Minimum Provision 4: Include Appropriate Monitoring at the Right Temporal and Spatial Scales

Hydrologic and ecological monitoring at the appropriate temporal and spatial scales is essential to provide groundwater managers with timely information to maintain sustainable yield (Gleeson et al. 2012), enable learning from consequences of management actions (Slattery 2016), and reduce the risk of making the wrong management or policy choice (Kriebel et al. 2001; Craig and Ruhl 2014). Monitoring locations with data collected at appropriate frequencies are needed in places that will provide an early warning if impacts are headed toward target resources (Harrington et al. 2017; Noordujin et al. 2019). A groundwater model that predicts where impacts are likely to occur can be used to help site those locations (Harrington et al. 2017).

As noted in Minimum Provision 2, monitoring of the ecological target alone or as the principal criterion for invoking action is not an effective strategy, especially because there is usually a lag between changed groundwater levels and ecosystem condition (Eamus et al. 2006). Once the ecological target shows impacts, it may be too late to take effective management actions because of the relatively slow movement of groundwater and the time that is needed for the ecological target to recover from the impact (Eamus et al. 2006; Leshy 2008; Barlow and Leake 2012). Thus, once the ecological targets are defined, it is important to select indicators that are predicted to be affected well before the targets themselves would be affected (Noordujin et al. 2019). Rohde et al. (2018) recommend that, especially for high-risk ecological targets, ecosystem impacts be monitored through a variety of indicators to make sure hydrologic results are true and on track to meeting management goals for natural systems. For example, to monitor sustainable groundwater use for the Sierra Vista Subwatershed in southeastern Arizona, the USPP uses 14 indicators that include groundwater levels and gradients, groundwater

budgets, and river and spring water quality and quantity measurements (Gungle et al. 2016).

Minimum Provision 5: Provide Accessible and Timely Reporting of Data

It is important that data be transparent, public, and shared in a timely manner so that appropriate mitigation actions in response to triggers can be taken and third parties can review and comment on results (Gleeson et al. 2012; Groundwater Governance Project 2016; Slattery 2016; Bureau of Meteorology 2017), particularly because impacts to natural resources from groundwater withdrawals could continue to worsen and take a long time to recover (Barlow and Leake 2012; Noordujin et al. 2019). Having data accessible to all stakeholders can reduce conflict and confusion and help to inform management discussions and decisions (Christian-Smith and Abhold 2015) and can also enable public participation in monitoring (Craig and Ruhl 2014). Useful information to include in reports include withdrawal and recharge amounts and timing, water levels, climatic trends, and assessments of likely effects on natural systems (Gungle et al. 2016).

All data are not necessarily good or appropriate data (Kirchner 2006), so it is also important to have objective technical review and quality control of the design of the monitoring regime, the resulting data, and any associated analysis and interpretation. In California, some groundwater sustainability agencies have created advisory and ad hoc committees to integrate a broad spectrum of interests and expertise, including biological and hydrologic, during the sustainable groundwater management planning process (e.g., Fox Canyon Groundwater Management Agency 2020; Santa Cruz Mid-County Groundwater Agency 2020; Sonoma Valley Groundwater Sustainability Agency 2020). Invited experts can offer neutral and objective review of data, models, and plans, which can provide a critical check-and-balance of vested interests. To better ensure objectivity, the method of appointing or selecting experts should not be left to any single stakeholder interest (Slattery 2016).

Minimum Provision 6: Implement Effective Management Actions if Triggers Are Reached

Along with clearly defined thresholds and triggers, protecting natural systems requires clear procedures and management actions to address situations where triggers indicate that existing actions are failing to meet management goals, including “stopping” rules (i.e., ceasing withdrawals) or “changing” rules that can halt or refine the action (Gregory et al. 2006). Predictive modeling can be helpful in determining appropriate management actions (Harrington et al. 2017; Wolaver et al. 2020). We note that actions that only call for “further study” (e.g., to review data, conduct additional monitoring, or to develop management or mitigation

measures) are insufficient as prescribed actions because they involve no change to water uses if a trigger is reached and therefore are not rigorously protective of natural systems (Slattery 2016; Thomann et al. 2020).

If impacts to natural systems cannot be avoided, minimized or mitigated, modification to the management framework or project plan may be required to ensure environmental sustainability. The preferred option is avoidance because minimization and mitigation often include risks of failure to meet conservation objectives (McKenney and Wilkinson 2015). Where specific management actions are proposed as part of planning (e.g., to minimize or mitigate potential impacts), the actions should be modeled before approval of proposed water uses (see Minimum Provision 4), and should only move forward if they are found to be consistent with achieving management objectives and sustainability goals.

Reducing, or even ceasing entirely, groundwater withdrawals may be the only means of responding to declining groundwater trends and associated triggers in some situations (Wolaver et al. 2020). Other options for groundwater sustainability may include changing the location and seasonality of pumping, offsetting proposed pumping by retiring existing water use, adopting water efficiency measures, crop switching, changing land use, and increasing managed aquifer recharge (Upper San Pedro Partnership 2013; Babbitt et al. 2018; Thomann et al. 2020). Retirement of active long-standing pumping from the same aquifer has been demonstrated by models and application as an effective offset or mitigation approach for groundwater withdrawals if the groundwater system of concern benefits at appropriate spatial and temporal scales (Lacher et al. 2014). For conflicts with existing human water uses, replacing reduced or lost groundwater with water from another source (e.g., treated wastewater effluent, stormwater, surface water, or groundwater from elsewhere) may be a viable action. Quantification of the amount of source water required for aquifer recharge projects may require the use of surface water models or monitoring data in conjunction with groundwater models (Lacher et al. 2014; Richter et al. 2014).

Caution is needed when mitigating lost groundwater with water from a different source (i.e., replacement water), especially for springs. Many springs contain endemic species (i.e., species found nowhere else) that are adapted to the unique characteristics of the specific water resources they use or inhabit (Davis et al. 2017). These characteristics, such as seasonal variability, water quality, temperature and other parameters often vary from one place to another (Sada et al. 2005; Abele 2011). Replacing one source of water with another with the same geochemistry and physical characteristics may be impossible in these situations. As demonstrated at Mound Springs in Australia in which re-injection of groundwater failed to restore spring vents and associated wetland areas, artificially replenishing water for a spring after its source groundwater has been drained off is very unlikely to restore sensitive species and ecosystems (Mudd 2000).

Withdrawals that may decrease spring flows should thus be avoided if the only mitigation measure is to augment flows with replacement water.

Sustainable water management involves tradeoffs across an array of social, economic, and environmental interests in terms of leaving water in place versus extracting it (Gungle et al. 2016). Once infrastructure is built, continuing water use may be driven by financing and other obligations which may limit flexibility in reducing or eliminating water use to prevent adverse impacts to natural systems. Thus, we strongly recommend adopting transparent decisional processes for planning and management that either involve reaching consensus among stakeholders, or allow an objective, third-party entity to choose management actions in furtherance of well-defined sustainability goals.

Minimum Provision 7: Secure Adequate Funding and Capacity for Project Planning, Implementation, Monitoring, and Reporting

Ensuring that natural systems will be adequately protected given historic and current groundwater withdrawals as well as future proposed water use necessitates an upfront and ongoing investment of time and money. In the case of projects, such investment is needed not only during project conceptualization, design, and approval, but for the duration of the project (Craig and Ruhl 2014; Poff et al. 2016). These time scales may span longer than the working lives of those associated with a proposed project at its outset (Walters and Holling 1990). The opportunity costs of foregone options, including risks of failure to protect natural systems, should be factored into project costs and budgets (Gregory et al. 2006).

Clarity about who is responsible for monitoring, reporting, and implementation costs is key, and should include what funding sources or mechanisms will be used. Verification that funds are available, have been set aside, or are accounted for in bonding mechanisms, can increase incentives and certainty in the implementation of sustainable groundwater management (Cody et al. 2015; Newman et al. 2018; Fletcher et al. 2019). Some mechanisms for generating funds include administrative, water use, or replenishment fees for water districts (Newman et al. 2018) as well as seeking assistance from federal programs like the U.S. Department of Agriculture's Conservation Reserve Enhancement Program (Cody et al. 2015). For agencies, fee-based approaches to fund monitoring, reporting, and management may reduce uncertainties of future agency budgets to maintain the minimum provisions, and fees may be structured to provide incentives to benefit groundwater sustainability (Nie and Schultz 2012; Newman et al. 2018).

By engaging diverse partners in the development of groundwater management efforts, a broader array of funding sources may become available and accessible for the design and construction of mitigation projects. For example, stormwater recharge projects that provide

recharge at a location that benefits groundwater-dependent ecosystems may also result in other multiple benefits such as erosion control or flood control, allowing for additional funding opportunities. The cost of effluent recharge projects can also be partially borne by the municipal entities who operate wastewater treatment facilities. Conservation easements can be a useful tool to permanently retire historic pumping or preclude future pumping in areas where groundwater withdrawals will most directly impact natural systems, and funds supporting wildlife habitat protection projects may be appropriate. Walters (2007) suggests that collaborative monitoring programs, incentives to promote sharing of data, and remote sensing and other monitoring innovations may help reduce monitoring costs, as has been seen in the Sierra Vista Subwatershed collaborative effort in Arizona (Upper San Pedro Partnership 2013).

Implementing Minimum Provisions 1 to 6 can be costly in terms of time, effort, and resources. Minimum Provision 7 is therefore essential to ensure the provisions will be implemented over the near and long terms.

Discussion and Conclusion

The magnitude of groundwater depletion is difficult to appreciate given its invisible nature. Konikow (2015) found that groundwater depletion incurred from 1900 to 2008 across the United States was about 1000 km³, which manifests as water table declines with costly impacts to human communities and natural systems. Large reductions in shallow groundwater have been documented across the western United States (Perrone and Jasechko 2017). In arid systems, groundwater is an essential water source for animal populations for many kilometers around desert streams and rivers, and rare and sensitive species associated with these water bodies may not be able to persist elsewhere (Sada et al. 2005; Hjort et al. 2015; Wolaver et al. 2020). Even slight drops in the water table of these natural systems can have irreversible impacts (Glazer and Likens 2012). Given that sensitivity, the physical complexities of many groundwater basins, the paucity of data required to understand groundwater dynamics and its connections to ecosystems, and the magnitude of many existing and proposed uses, the minimum provisions described here are necessary baseline measures to reduce the risk of adverse impacts to natural systems from historic, current, and future groundwater use.

These minimum provisions have similarities to steps in adaptive management, a conservation approach that can be used to provide flexibility under uncertainty for natural systems (Nie and Schultz 2012). The core principle of adaptive management is the establishment of a feedback cycle wherein a natural system is monitored and management is adjusted based on observation and trends (Nie and Schultz 2012; Thomann et al. 2020). Thomann et al. (2020) reviewed 11 case studies to examine the application of adaptive management to groundwater planning and management. They found significant shortcomings of adaptive management as

applied to these systems, particularly with regard to poorly defined mitigation measures to address adverse environmental impacts.

Allen and Gunderson (2011) describe a “sweet spot” for successful application of adaptive management: it is a problem context with (1) high uncertainty, (2) high controllability (i.e., decision-makers have greater capacity to intervene in the management problem), and (3) low risk (i.e., chance that interventions can lead to irreversible adverse consequences is low). Groundwater management clearly meets the first criterion and may meet the second in some situations (e.g., California groundwater basins covered by the Sustainable Groundwater Management Act). However, the sensitivity of natural systems to reduced access to groundwater, especially in arid regions, creates a high-risk situation. Craig and Ruhl (2014) argue that more controllability and risk reduction could be achieved with legislation, and they have proposed wording of such legislation. On the other hand, Molle and Closas (2020) note that regulations are never sufficient without a willingness to deploy regulatory power, which has been a shortfall of state-centered groundwater governance. Because groundwater can be overused or rendered unusable through mismanagement, sustainable groundwater governance requires robust political processes that include stakeholders representing different economic and social sectors and the environment (Robertson 2020).

Aquifers where relatively small volumes of groundwater have been removed from storage, and where no large volume groundwater extraction is likely in the future, seem the best candidates for adaptively managing groundwater to meet sustainability goals, given the more manageable level of uncertainty of these natural systems. Large volume water development projects have nevertheless been proposed and implemented in sensitive desert systems with low resiliency (Parker et al. 2021). Such projects introduce considerable additional uncertainty with much greater risk of adverse or irreversible impacts for natural systems over the long term. Adherence to all of the minimum provisions described in this paper is particularly crucial to reduce the risk of deleterious impacts to these systems. Moreover, precaution would dictate that if new groundwater uses are allowed, they be smaller from the outset and then monitored to affirm no unacceptable impact (Fletcher et al. 2019) before allowing use to ratchet up, rather than approving larger initial uses based on the supposition they could be curtailed later if need be. Such an approach would also provide an opportunity to test and verify the adequacy of the monitoring regime and adaptive management mechanisms necessary to achieve sustainability goals, while also reducing the risk of over-building water infrastructure and incurring extra implementation costs (Fletcher et al. 2019).

Finally, we underscore that while these minimum provisions are necessary, they might not be sufficient to protect natural systems in all cases. Thus, we encourage further multidisciplinary study of the performance of different management regimes with regard to how they deliver environmental protections so as to adaptively

improve future guidance for practitioners. We also note that while sustainable groundwater management requires balancing social, economic and environmental values, elucidation of best practices for making such tradeoff decisions was beyond the scope of this paper. That too warrants scientific examination and synthesis for practitioners. Importantly, the minimum provisions presented here not only help reduce scientific knowledge gaps, they also can help planners and managers identify meaningful points of engagement and accountability with stakeholders. Applied in concert, the provisions described herein improve the likelihood that groundwater use and development will meet sustainability goals over the near and long terms and avoid adverse impacts to groundwater-dependent ecosystems and species.

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References

- Abele, S. (Ed). 2011. *Nevada Springs Conservation Plan*. Reno: The Nature Conservancy. <http://heritage.nv.gov/node/171> (accessed January 17, 2020)
- Allen, C.R., and L.H. Gunderson. 2011. Pathology and failure in the design and implementation of adaptive management. *Journal of Environmental Management* 92: 1379–1384.
- Alley, W.M., and S.A. Leake. 2004. The journey from safe yield to sustainability. *Groundwater* 42, no. 1: 12–16.
- Alley, W.M., T.E. Reilly, and O.L. Franke. 1999. Sustainability of ground-water resources. U.S. Geological Survey Circular 1186. Denver, Colorado: U.S. Geological Survey. <https://pubs.usgs.gov/circ/1999/circ1186/pdf/circ1186.pdf> (accessed June 24, 2019).
- Arizona Department of Water Resources. 2021. Active management areas. <https://new.azwater.gov/ama> (accessed January 27, 2021).
- Babbitt, C., K. Gibson, S. Sellers, N. Brozovic, A. Saracino, A. Hayden, M. Hall, and S. Zellmer. 2018. *The Future of Groundwater in California: Lessons in Sustainable Management from across the West*. New York: Environmental Defense Fund. <https://www.edf.org/ecosystems/future-groundwater-california> (accessed October 26, 2020)
- Barlow, P.M., and S.A. Leake. 2012. Streamflow depletion by wells—Understanding and managing the effects of groundwater pumping on streamflow. U.S. Geological Survey Circular 1376. Reston, Virginia: U.S. Geological Survey. <http://pubs.usgs.gov/circ/1376> (accessed September 4, 2018).
- Bredehoeft, J.D. 2002. The water budget myth revisited: Why hydrogeologists model. *Groundwater* 40, no. 4: 340–345.
- Bredehoeft, J.D., S.S. Papadopoulos, and H.H. Cooper. 1982. Groundwater: The water-budget myth. In National Research Council. *Scientific Basis of Water-Resource Management*, 51–57. Washington, DC: The National Academies Press.
- Bureau of Meteorology. 2017. *Good Practice Guidelines for Water Data Management Policy: World Water Data Initiative*. Melbourne: Bureau of Meteorology. <http://www.bom.gov.au/water/about/publications/document/Good-Practice-Guidelines-for-Water-Data-Management-Policy.pdf> (accessed December 14, 2019)
- Cantarelli, C.C., B. Flyvbjerg, B. van Wee, and E.J.E. Molin. 2010. Lock-in and its influence on the project performance of large-scale transportation infrastructure projects: Investigating the way in which lock-in can emerge and affect cost overruns. *Environment and Planning B: Planning and Design* 37: 792–807. <https://doi.org/10.1068/b36017>
- Christian-Smith, J., and K. Abhold. 2015. *Measuring What Matters: Setting Measurable Objectives to Achieve Sustainable Groundwater Management in California*. Oakland: Union of Concerned Scientists <https://www.ucsusa.org/global-warming/regional-information/california-and-western-states/measuring-what-matters> (accessed June 24, 2019)
- Cody, K.C., S.M. Smith, M. Cox, and K. Andersson. 2015. Emergence of collective action in a groundwater commons: Irrigators in the San Luis Valley of Colorado. *Society and Natural Resources* 28: 405–422. <https://doi.org/10.1080/08941920.2014.970736>
- Craig, R.K., and J.B. Ruhl. 2014. Designing administrative law for adaptive management. *Vanderbilt Law Review* 67, no. 1: 1–88.
- Crawford-Brown, D., and S. Crawford-Brown. 2011. The precautionary principle in environmental regulations for drinking water. *Environmental Science and Policy* 14: 379–387.
- Crevier, L.P., and L. Parrott. 2019. Synergy between adaptive management and participatory modelling: The two processes as interconnected spirals. *Ecological Informatics* 53: 100982. <https://doi.org/10.1016/j.ecoinf.2019.100982>
- Davis, J.A., A. Kerecsy, and S. Nicol. 2017. Springs: Conserving perennial water is critical in arid landscapes. *Biological Conservation* 211: 30–35. <https://doi.org/10.1016/j.biocon.2016.12.036>
- Deacon, J.E., A.E. Williams, C.D. Williams, and J.E. Williams. 2007. Fueling population growth in Las Vegas: How large-scale groundwater withdrawal could burn regional biodiversity. *Bioscience* 57, no. 8: 689–698.
- de Graaf, I.E.M., T. Gleeson, L.P.H. van Beek, E.H. Sutanudjaja, and M.F.P. Bierkens. 2019. Environmental flow limits to global groundwater pumping. *Nature* 574: 90–94. <https://doi.org/10.1038/s41586-019-1594-4>
- Dieter, C.A., M.A. Maupin, R.R. Caldwell, M.A. Harris, T.I. Ivahnenko, J.K. Lovelace, N.L. Barber, and K.S. Linsey. 2018. *Estimated Use of Water in the United States in 2015*. U.S. Geological Survey Circular 1441. Reston, Virginia: U.S. Geological Survey. 65 p. <https://doi.org/10.3133/cir1441>
- Eamus, D., R. Froend, R. Loomes, G. Hose, and B. Murray. 2006. A functional methodology for determining the groundwater regime needed to maintain the health of groundwater-dependent vegetation. *Australian Journal of Botany* 54: 97–114.
- Fischman, R.L., and J.B. Ruhl. 2016. Judging adaptive management practices of U.S. agencies. *Conservation Biology* 30, no. 2: 268–275. <https://doi.org/10.1111/cobi.12616>

- Fletcher, S., K. Strzepek, A. Alsaati, and O. deWeck. 2019. Learning and flexibility for water supply infrastructure planning under groundwater resource uncertainty. *Environmental Research Letters* 14: 114022. <https://doi.org/10.1088/1748-9326/ab4664>
- Flyvbjerg, B. 2014. What you should know about megaprojects and why: An overview. *Project Management Journal* 45, no. 2: 6–19. <https://doi.org/10.1002/pmj.21409>
- Fox Canyon Groundwater Management Agency. 2020. *Fox canyon groundwater facilitated process*. <http://fcgmasustainability.org/overview/> (accessed October 26, 2020)..
- Freed, Z., A. Aldous, and M.W. Gannett. 2019. Landscape controls on the distribution and ecohydrology of central Oregon springs. *Ecohydrology* 12, no. 2: e2065. <https://doi.org/10.1002/eco2065>
- Gillson, L., H. Biggs, I.P.J. Smit, M. Virah-Sawmy, and K. Rogers. 2019. Finding common ground between adaptive management and evidence-based approaches to biodiversity conservation. *Trends in Ecology and Evolution* 34, no. 1: 31–44.
- Glazer, A.N., and G.E. Likens. 2012. The water table: The shifting foundation of life on land. *Ambio* 41: 657–669. <https://doi.org/10.1007/s13280-012-0328-8>
- Gleeson, T., W.M. Alley, D.M. Allen, M.A. Sophocleous, Y. Zhou, M. Taniguchi, and J. VanderSteen. 2012. Towards sustainable groundwater use: Setting long-term goals, back-casting, and managing adaptively. *Groundwater* 50, no. 1: 19–26. <https://doi.org/10.1111/j.1745-6584.2011.00825x>
- Grantham, T.E., and J.H. Viers. 2014. 100 years of California's water rights system: Patterns, trends and uncertainty. *Environmental Research Letters* 9: 084012. <https://doi.org/10.1088/1748-9326/9/8/084012>
- Gregory, R., D. Ohlson, and J. Arvai. 2006. Deconstructing adaptive management: Criteria for applications to environmental management. *Ecological Applications* 16, no. 6: 2411–2425.
- Groffman, P.M., J.S. Baron, T. Blett, A.J. Gold, I. Goodman, L.H. Gunderson, B.M. Levinson, M.A. Palmer, H.W. Paerl, G.D. Peterson, N.L. Poff, D.W. Rejeski, J.F. Reynolds, M.G. Turner, K.C. Weathers, and J. Wiens. 2006. Ecological thresholds: The key to successful environmental management or an important concept with no practical application? *Ecosystems* 9: 1–13. <https://doi.org/10.1007/s10021-003-0142-z>
- Groundwater Governance Project. 2016. *Global Diagnostic on Groundwater Governance*. Rome, Italy: GEF, UNESCO-IHP, FAO, World Bank and IAH. Available at <http://www.fao.org/3/a-i5706e.pdf> (accessed December 20, 2019).
- Groves, D.G., E. Molina-Perez, E. Bloom, and J.R. Fischbach. 2019. Robust decision making (RDM): Applications to water planning and climate policy. In *Decision Making under Deep Uncertainty*, ed. V. Marchau, W. Walker, P. Bloemen, and S. Popper, 135–162. Cham: Springer. https://doi.org/10.1007/978-3-030-05252-2_7
- Gungle, B., J.B. Callegary, N.V. Paretti, J.R. Kennedy, C.J. Eastoe, D.S. Turner, J.E. Dickinson, L.R. Levick, and Z.P. Sugg. 2016. Hydrological conditions and evaluation of sustainable groundwater use in the Sierra Vista Subwatershed, Upper San Pedro Basin, southeastern Arizona (ver.1.2, February 2017). U.S. Geological Survey Scientific Investigations Report 2016-5114, Reston, Virginia: U.S. Geological Survey. 90 p. <https://doi.org/10.3133/sir20165114>
- Harrington, R., K. Rainville, and T.N. Blandford. 2017. Comment on “drawdown ‘triggers’: A misguided strategy for protecting groundwater-fed streams and springs” by Matthew J. Currell, 2016, v.54, no. 5: 619–612. *Groundwater* 55, no. 2: 152–153.
- Hjort, J., J.E. Gordon, M. Gray, and M.L. Hunter Jr. 2015. Why geodiversity matters in valuing nature's stage. *Conservation Biology* 29, no. 3: 630–639. <https://doi.org/10.1111/cobil.12510>
- Kalf, F.P., and D.R. Wooley. 2005. Applicability and methodology of determining sustainable yield in groundwater systems. *Hydrogeology Journal* 13: 295–312. <https://doi.org/10.1007/s10040-004-0401-x>
- Kath, J., A.J. Boulton, E.T. Harrison, and F.J. Dyer. 2018. A conceptual framework for ecological responses to groundwater regime alteration (FERGRA). *Ecohydrology* 11: e2010. <https://doi.org/10.1002/eco.2010>
- Kendy, E., C. Apse, and K. Blann. 2012. *A practical guide to environmental flows for policy and planning*. The Nature Conservancy. 72 p. <https://www.conservationgateway.org/ConservationPractices/Freshwater/EnvironmentalFlows/MethodsandTools/ELOHA/Pages/ecological-limits-hydrolo.aspx> (accessed August 11, 2020).
- Kiesecker, J.M., H. Copeland, A. Pocerwicz, and B. McKenney. 2010. Development by design: Blending landscape-level planning within the mitigation hierarchy. *Frontiers in Ecology and the Environment* 8, no. 5: 261–266. <https://doi.org/10.1890/090005>
- Kirchner, J.W. 2006. Getting the right answers for the right reasons: Linking measurements, analyses, and models to advance the science of hydrology. *Water Resources Research* 42: W03S04. <https://doi.org/10.1029/2005WR004362>
- Kodric-Brown, A., and J.H. Brown. 2007. Native fishes, exotic mammals, and the conservation of desert springs. *Frontiers in Ecology and the Environment* 5, no. 10: 549–553. <https://doi.org/10.1890/070002>
- Konikow, L.F. 2015. Long-term groundwater depletion in the United States. *Groundwater* 53, no. 1: 2–9. <https://doi.org/10.1111/gwat.12306>
- Kriebel, D., J. Tickner, P. Epstein, J. Lemons, R. Levins, E.L. Loechler, M. Quinn, R. Rudel, T. Schettler, and M. Stoto. 2001. The precautionary principle in environmental science. *Environmental Health Perspectives* 109: 871–876.
- Lacher, L., D. Turner, B. Gungle, B. Bushman, and H. Richter. 2014. Application of hydrologic tools and monitoring to support managed aquifer recharge decision making in the Upper San Pedro River, Arizona, USA. *Water* 6, no. 11: 3495–3527. <https://doi.org/10.3390/w6113495>
- Leshy, J.D. 2008. Interstate groundwater resources: The federal role. *Hastings West-Northwest Journal of Environmental Law and Policy* 1475. http://repository.uchastings.edu/faculty_scholarship/369 (accessed December 15, 2019).
- McCarl, B.A., C.R. Dillon, K.O. Keplinger, and R.L. Williams. 1999. Limiting pumping from the Edwards Aquifer: An economic investigation of proposals, water markets, and spring flow guarantees. *Water Resources Research* 35, no. 4: 1257–1268.
- McKenney, B., and J. Wilkinson. 2015. *Achieving Conservation and Development: 10 Principles for Applying the Mitigation Hierarchy*. 20. Arlington, Virginia: The Nature Conservancy. <https://www.conservationgateway.org/Documents/TNCApplyingTheMitigationHierarchy.pdf> (accessed June 24, 2019).
- Meadows, D. 1998. *Indicators and Information Systems for Development*. Hartford: The Sustainability Institute.
- Megdal, S.B., A.K. Gerlak, R.G. Varady, and L.-Y. Huang. 2015. Groundwater governance in the United States: Common priorities and challenges. *Groundwater* 53, no. 5: 677–684. <https://doi.org/10.1111/gwa.12294>
- Merritt, D.M., and H.L. Bateman. 2012. Linking stream flow and groundwater to avian habitat in a desert riparian system. *Ecological Applications* 22, no. 7: 1973–1988.
- Molle, F., and A. Closas. 2020. Why is state-centered groundwater governance largely ineffective? A review. *WIREs Water* 7, no. 1: e1935. <https://doi.org/10.1002/wat2.1395>
- Moritz, M.A., M.D. Hurteau, K.N. Suding, and C.M. D'Antonio. 2013. Bounded ranges of variation as a framework for

- future conservation and fire management. *Annals of the New York Academy of Sciences* 1286: 92–107.
- Mudd, G.M. 2000. Mound springs of the Great Artesian Basin in South Australia: A case study from Olympic Dam. *Environmental Geology* 39, no. 5: 463–476.
- Museum of Northern Arizona Springs Stewardship Institute. 2020. *Conservation Strategy for Springsnails in Nevada and Utah. Version 1.0*. Reno: Nevada Department of Wildlife and Salt Lake City: Utah Division of Wildlife Resources. http://docs.springstewardship.org/Springsnails/SpringsnailFinal_26_Aug_2020_Final.pdf (accessed November 11, 20)
- Naumburg, E., R. Mata-Gonzalez, R.G. Hunter, T. McLendon, and D.W. Martin. 2005. Phreatophytic vegetation and groundwater fluctuations: A review of current research and application of ecosystem response modeling with an emphasis on Great Basin vegetation. *Environmental Management* 35, no. 6: 726–740.
- Nelson, R., and P. Ouevaullier. 2016. Groundwater law. In *Integrated Groundwater Management: Concepts, Approaches and Challenges*, ed. A.J. Jakeman, O. Barreteau, R.J. Hunt, J.-D. Rinaudo, and A. Ross, 173–196. SpringerOpen. <https://doi.org/10.1007/978-3-319-23576-9>. <https://link.springer.com/book/10.1007%2F978-3-319-23576-9> (accessed January 2, 2021)
- Newman, C., R. Howitt, and D. MacEwan. 2018. How are Western water districts managing groundwater basins? *California Agriculture* 72, no. 1: 28–37. <https://doi.org/10.3733/ca.2018a0004>. <http://calag.ucanr.edu/Archive/?article=ca.2018a0004> (accessed August 9, 2020)
- Nie, M.A., and C.A. Schultz. 2012. Decision-making triggers in adaptive management. *Conservation Biology* 26, no. 6: 1137–1144. <https://doi.org/10.1111/j.1523-1739.2102.01915.x>
- Noorduijn, S.L., P.G. Cook, C.T. Simmons, and S.B. Richardson. 2019. Protecting groundwater levels and ecosystems with simple management approaches. *Hydrogeology Journal* 27: 225–237. <https://doi.org/10.1007/s10040-018-1849-4>
- Orellana, F., P. Verma, S.P. Loheide II., and E. Daly. 2012. Monitoring and modeling water-vegetation interactions in groundwater-dependent ecosystems. *Reviews of Geophysics* 50: RG3003. <https://doi.org/10.1029/2011RG000383>
- Ostrom, E., M.A. Janssen, and J.M. Anderies. 2007. Going beyond panaceas. *Proceedings of the National Academy of Sciences of the United States of America* 104, no. 39: 15176–15178. <https://doi.org/10.1073/pnas.0701886104>
- Parker, S.S., A. Zdon, W.T. Christian, B.S. Cohen, M.P. Mejia, N.S. Frag, E.E. Curd, K. Edalati, and M.A. Renshaw. 2021. Conservation of Mojave Desert springs and associated biota: Status, threats, and policy opportunities. *Biodiversity and Conservation* 30: 311–327. <https://doi.org/10.1007/s10531-020-02090-7>
- Patten, D.T., L. Rouse, and J.C. Stromberg. 2008. Isolated spring wetlands in the Great Basin and Mojave Deserts, USA: Potential response of vegetation to groundwater withdrawal. *Environmental Management* 41: 398–413. <https://doi.org/10.1007/s00267-007-9035-9>
- Perrone, D., and S. Jasechko. 2017. Dry groundwater wells in the western United States. *Environmental Research Letters* 12: 104002. <https://doi.org/10.1088/1748-9326/aa8ac0>
- Poff, N.L., C.M. Brown, T.E. Grantham, J.H. Matthews, M.A. Palmer, C.M. Spence, R.L. Wilby, M. Haasnoot, G.F. Mendoza, K.C. Dominique, and A. Baeza. 2016. Sustainable water management under future uncertainty with eco-engineering decision scaling. *Nature Climate Change* 6: 25–34.
- Pool, D.R., and J.E. Dickinson. 2007. Ground-water flow model of the Sierra Vista Subwatershed and Sonoran portions of the Upper San Pedro Basin, southeastern Arizona, United States, and northern Sonora, Mexico. Reston: U.S. Geological Survey Scientific Investigations Report 2006-5228. <https://pubs.usgs.gov/sir/2006/5228/> (accessed October 27, 2020).
- Rains, M.C., J.F. Mount, and E.W. Larsen. 2004. Simulated changes in groundwater and vegetation distributions under different reservoir operations scenarios. *Ecological Applications* 14, no. 1: 192–207.
- Reilly, T.E., K.F. Dennehy, W.M. Alley, and W.L. Cunningham. 2008. *Ground-Water Availability in the United States*, 70. Reston: U.S. Geological Survey: U.S. Geological Survey Circular 1323. <https://pubs.usgs.gov/circ/1323/> (accessed August 12, 2020)
- Richter, H.E., B. Gungle, L.J. Lacher, D.S. Turner, and B.M. Bushman. 2014. Developing a shared vision for groundwater management to protect and sustain baseflows of the Upper San Pedro River, Arizona, USA. *Water* 6: 2519–2538. <https://doi.org/10.3390/w6082519>
- Robertson, J. 2020. Challenges in sustainably managing groundwater in the Australian Great Artesian Basin: Lessons from current and historic legislative regimes. *Hydrogeology Journal* 28: 343–360. <https://doi.org/10.1007/s10040-019-02047-3>
- Rohde, M.M., L. Saito, and R. Smith. 2020. *Groundwater Thresholds for Ecosystems: A Guide for Practitioners*. Santa Cruz: The Nature Conservancy. <https://groundwaterresourcehub.org/sgma-tools/> (accessed November 25, 2020).
- Rohde, M.M., S.B. Sweet, C. Ulrich, and J. Howard. 2019. A transdisciplinary approach to characterize hydrological controls on groundwater-dependent ecosystem health. *Frontiers in Environmental Science* 7: 175. <https://doi.org/10.3389/fenvs.2019.00175>
- Rohde, M.M., S. Matsumoto, J. Howard, S. Liu, L. Riege, and E.J. Remson. 2018. *Groundwater Dependent Ecosystems under the Sustainable Groundwater Management Act: Guidance for Preparing Groundwater Sustainability Plans*. San Francisco: The Nature Conservancy. <https://groundwaterresourcehub.org/sgma-tools/gsp-guidance-document/> (accessed November 14, 2020)
- Rohde, M.M., R. Froend, and J. Howard. 2017. A global synthesis of managing groundwater dependent ecosystems under sustainable groundwater policy. *Groundwater* 55, no. 3: 293–301.
- Rudestam, K., and R. Langridge. 2014. Sustainable yield in theory and practice: Bridging scientific and mainstream vernacular. *Groundwater* 52, no. S1: 90–99. <https://doi.org/10.1111/gwat.12160>
- Sada, D.W., E. Fleischman, and D.D. Murphy. 2005. Associations among spring-dependent aquatic assemblages and environmental and land use gradients in a Mojave Desert mountain range. *Diversity and Distributions* 11: 91–99.
- Santa Cruz Mid-County Groundwater Agency. 2020. Surface water working group. <https://www.midcountygroudwater.org/gsp-advisory-committee/surface-water-working-group> (accessed October 26, 2020).
- Sizek, J. 2018. California's groundwater regime: The Cadiz case. *Resources* 7, no. 1: 7. <https://doi.org/10.3390/resources701007>
- Slattery, C. 2016. Canary in the coal mine: Why the approval conditions for the Carmichael mine reveal the need to amend the EPBC Act to include adaptive management principles. *Environmental and Planning Law Journal* 33, no. 5: 421–442.
- Sonoma Valley Groundwater Sustainability Agency. 2020. Advisory committee. <http://sonomavalleygroundwater.org/ac/> (accessed October 26, 2020)
- Stromberg, J.C., J.A. Tress, S.D. Wilkins, and S.D. Clark. 1992. Response of velvet mesquite to groundwater decline. *Journal of Arid Environments* 23: 45–58.
- Thomann, J.A., A.D. Werner, D.J. Irvine, and M.J. Currell. 2020. Adaptive management in groundwater planning and

development: A review of theory and applications. *Journal of Hydrology* 586: 124871. <https://doi.org/10.1016/j.jhydrol.2020.124871>

Upper San Pedro Partnership. 2013. *Water management of the regional aquifer in the Sierra Vista Subwatershed, Arizona—2011 report to Congress*, 16. Washington, D.C.: Department of the Interior. <http://uppersanpedropartnership.org/wp-content/uploads/2017/10/2011321ReportDRAFT05-07-13.pdf> (accessed April 17, 2020)

Walters, C.J. 2007. Is adaptive management helping to solve fisheries problems? *Ambio* 36, no. 4: 304–307.

Walters, C.J., and C.S. Holling. 1990. Large-scale management experiments and learning by doing. *Ecology* 71, no. 6: 2060–2068.

Wolaver, B.D., S.C. Priestley, L.J. Crossey, K.E. Karlstrom, and A.J. Love. 2020. Elucidating sources to aridland Dalhousie Springs in the Great Artesian Basin (Australia) to inform

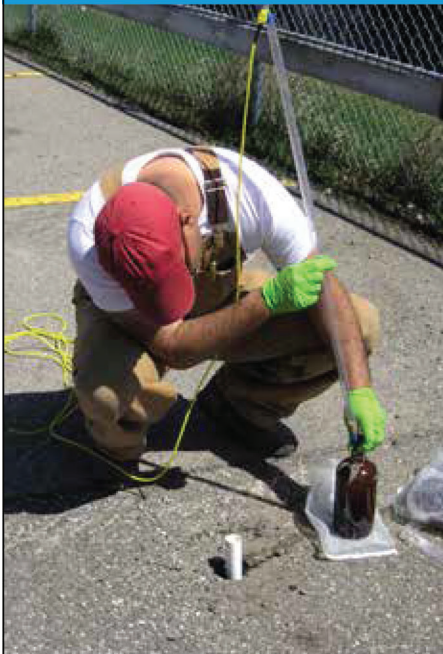
conservation. *Hydrogeology Journal* 28: 279–296. <https://doi.org/10.1007/s10040-019-02072-2>

Zdon, A., M.L. Davisson, and A.H. Love. 2018. Understanding the source of water for selected springs within Mojave Trails National Monument, California. *Environmental Forensics* 19, no. 2: 99–111. <https://doi.org/10.1080/15275922.2018.14-48909>

Zhou, Y. 2009. A critical review of groundwater budget myth, safe yield and sustainability. *Journal of Hydrology* 370: 207–213.

Zipper, S.C., F. Jaramillo, L. Wang-Erlandsson, S.E. Cornell, T. Gleeson, M. Porkka, T. Hävhä, A.-S. Crépin, I. Fetzer, D. Gerten, H. Hoff, N. Matthews, C. Ricaurte-Villota, M. Kumm, Y. Wada, and L. Gordon. 2020. Integrating the water planetary boundary with water management from local to global scales. *Earth's Future* 8: e2019EF001377. <https://doi.org/10.1029/2019EF001377>

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